`CompositionMeasure for MultiDP`

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of `CompositionMeasure` for `MultiDP`.

1 Hoare Triple

Precondition

Compiler-verified

The output measure is derived from the common capabilities of all child measures. Only capabilities with a common composition theorem are retained.

Caller-verified

None.

Postcondition

Theorem 1.1. `composability` returns `Ok(out)` if the composition of a vector of privacy guarantees is bounded above by `compose(d_mids)` under the reported adaptivity and composability. Otherwise it returns an error.

Proof. The composition implementation evaluates each representation only when every child supplies that representation. Native zCDP parameters are composed by adding ρ_i and source deltas. Pure RDP parameters are composed pointwise by adding their curves; pure zCDP is embedded into RDP using the exact $\alpha\rho_i$ relation. Gaussian-DP parameters, when enabled, are composed by $\mu^2 = \sum_i \mu_i^2$.

Each accumulator uses directed interval arithmetic, so its returned value is an upper bound on the corresponding exact composition. Approximate profiles and tradeoff functions are not composed without a representation-specific theorem. If no common supported capability exists, construction fails rather than advertising an unsupported composition. Therefore every representation retained in the resulting `PrivacyGuarantee` is a valid bound for the same composed privacy relation. □

References